

Part 6

Build, test and publish

The TF3 iteration loop is shorter than TSC's because the Model Editor reloads on file change and validates on demand. This part sets up that loop, gives an in-game test order for rail vehicles, covers mod packaging for mod.io, and ends with a symptom table.

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Model Editor behaviour from its manual page; mod.json/modinfo.json details from the Repaint Mods page only (the Mod Definition and Publish pages were not supplied).

1 • The iteration loop

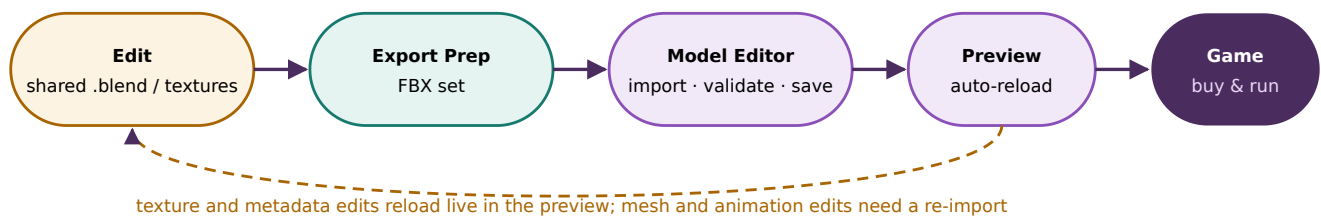


Figure 6.1 • Shorter than the TSC loop. With **Enable Auto Update** on, a saved texture or edited .mtl/.mdl reloads in the viewport without re-import. **manual**

You changed...	Redo
Textures	Save .tga into the mod's texture folder; the preview reloads (choose <i>On Focus</i> if you see black textures). Model Editor converts to .dds on save. manual
Material type or maps	Materials window, or re-import with Overwrite Materials off to keep edits. manual
Metadata values	Metadata window; Validate; Save (MODEL only). manual
Mesh geometry / hierarchy / names	Export Prep → drag the folder again with Overwrite Meshes on; metadata is kept if you leave Overwrite Materials/Animations off or re-apply the .import.lua. manual
Animation keys or strip names	Re-import with Overwrite Animations on. manual
LOD distances, bounding box	Model window (per-LOD tabs; recalculate bounding box / extent buttons). manual

2 • The mod skeleton

```
modinfo.json (display data for the mod
browser)
{
  "name": "My Loco",
  "description": "...",
  "authors": [ { "name": "you", "role":
"CREATOR" } ],
  "minorVersion": 0,
  "severityAdd": "NONE", "severityRemove":
"WARNING",
  "tags": [ "Train", "Vehicle" ]
}
```

Keys shown are the TF2 convention; the manual only states what the file is for (name, description, authors, preview image, dependency note). [check](#) against the Mod Definition page.

Create the mod folder in the staging area *before* the first import so the Model Editor can select it as the save target. [manual](#)

```
mod.json (technical definition)
{
  "dependencies": [ "someone_othermod_1" ],
  ...
}
```

“Most important is the specification of the correct dependency” when you reference another mod’s meshes/materials with `modid::`. [manual](#) Other fields [check](#).

Also: `strings.json` for the `_(“...”)` keys used in `description`, and the preview image.

3 • Testing in the Model Editor

#	Check (Options ► Rendering / Config)	Looks for
1	Axes + Nodes + Node Names on	Front along +x; node origins where you expect; names as in the metadata.
2	Bogies / Fake Bogies locators	Pivot markers midway between axles; fake bogie where a body lacks a second bogie.
3	Bounding Box, Extent, Collider	Box encloses the model (the <code> bounding_box</code> helper did its job); extent sensible.
4	LOD selector 0, 1, ... and the Lod scale slider	Every LOD complete; animated parts present at LOD 1.
5	Animation dropdown + loop	Each event strip plays, starts at rest, doors clear the body.
6	Seat Locators, Passengers/Cargo sliders	Crew in the cab, passengers in seats, nothing floating.
7	Light Source Locators, Time of Day at night	Headlights where the meshes are; emissive lamps glow.
8	Preview Particles with velocity	Smoke from the chimney, scaled by speed.
9	Color and Age sliders	Colour-blend mask recolours the right areas; dirt/rust layers plausible.
10	Validate button	No messages left (Part 4 §6).
11	Screenshot	Icons generated into the mod.

All options named exist in the manual's Options chapter. [manual](#)

4 • Testing in the game

- 1 Enable the mod for a new or existing game (mod manager). [check](#) menu names.
- 2 Buy the vehicle in a rail depot: it must appear in the store with its icons, name and description (needs `filterTags` matching the depot, default is fine, and the year within `availability`). [manual](#)
- 3 Couple a wagon and run a line: watch coupling distance, bogie tracking in curves, wheel spin rate, sagging or floating (axle origins).
- 4 Stop at a station: doors open on the platform side after `arrivalDelay`, passengers board, doors close before departure.
- 5 Reverse at a terminus: headlights/tail lamps switch ends; pantograph instance switches; crew seat moves to the leading end.
- 6 Night: lights on; **emissions** and **speed** in the vehicle window as intended.
- 7 Multiple unit and repaint variants appear grouped in the store if configured.

5 • Repaint and variant workflow

- Liveries: one `.mtl` per re-textured material, one `.mdl` per variant pointing at them in every LOD, new icons, `groupFileName` to group them. [manual](#)
- If you repaint someone else's vehicle, obtain permission, use a different modId and declare the dependency. [manual](#)
- Work from texture source files, not the shipped `.dds`. [manual](#)
- The Model Editor's **Copy** tool duplicates a model into another mod; **Rename** does search/replace on file names; **Convert** migrates TF2 models. [manual](#)

6 • Publishing

- TF3 mods are shared through **mod.io**; the manual's "Publish a mod" page was not supplied [check](#) its upload steps and the Curated Mods Program rules.
- Ship only what is yours and new: do not re-ship base-game or other mods' files; reference them with `::/` and `modid::/` instead. [manual](#)
- Include all three UI icons per model, `strings.json` entries for every `_("...")`, and a preview image.
- Run Validate on every model (Tools ► Validate handles many at once). [manual](#)

7 • Troubleshooting

Symptom	Cause → fix
Model Editor won't start / can't find mods	Config path with backslashes or wrong <code>userDataPath</code> → forward slashes, correct folder. manual

Symptom	Cause → fix
Imported model lies on its side or faces sideways	“Use Blender Coordinate System” unticked, or FBX exported with the wrong forward/up → X forward, Z up. manual
Node names end in .001 / meshes duplicated	Blender suffixes; use # comments or let Export Prep rename. manual
Black textures after saving in the paint program	Auto Update raced the file write → On Focus. manual
Material count mismatch warnings	Material slots vs mesh groups differ → fix slots in Blender, re-import. manual
Axles wobble	Axle origin not centred in x/z → fix origin. manual
Body misaligned in curves	Bogie origins not between axles, or a single-bogie body without a fake bogie. manual
Vehicle not in the depot	Year outside availability; filterTags mismatch; multipleUnitOnly = true . manual
Multiple unit missing	One part outside its availability years. manual
Doors / lights never animate	Strip name not an event; missing _on / _off pair; custom transformator not calling the util helpers.
Ship never reaches top speed	(Water only) area too large for availPower. manual
Crash or glitch on a skinned mesh	Skinned mesh with unskinned material → remove vertex groups or use Skin material. manual